

Vinh Ly

Palo Alto, CA | vily437@outlook.com | (318) 202-8037
<https://www.linkedin.com/in/vinh-ly-14b56b50>

Summary

Software Engineer with **18 years** of experience building and scaling distributed systems across Google, YouTube, Whatnot, and Microsoft. Led end-to-end ownership of developer platforms serving **hundreds of millions** of users, payment infrastructure processing **millions** of transactions, and internal security tooling adopted by **thousands** of engineers. Built and managed teams delivering high-traffic surfaces at Google including Chrome-infra and internal security platforms. Expert in **React, TypeScript, Python, Go**, and **distributed systems** design. Obsesses over system reliability, performance, and developer experience. On-call rotation ownership across all major roles. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Frontend: React, TypeScript, JavaScript, HTML, CSS, Next.js, GraphQL, REST APIs

Backend: Python, Go, C#, .NET, PostgreSQL, Redis, Kafka

Infrastructure: Kubernetes, GCP, AWS, Docker, Terraform, Bigtable, Spanner

Specializations: Distributed systems, System reliability, Payment systems, Security tooling, CI/CD pipelines, Performance optimization, Node.js

Work Experience

Staff Software Engineer, YouTube | 02/2025 - Present

- Own video platform infrastructure serving billions of daily requests, driving core initiatives to improve streaming reliability and reduce latency for global audience.
- Architect solutions for high-throughput content delivery systems, collaborating with cross-functional teams to optimize playback performance across device categories.
- Lead design reviews and technical direction for distributed subsystems handling real-time traffic spikes during major live events.

Staff Software Engineer, Whatnot | 05/2024 - 02/2025

- Owned payment processing pipeline handling **millions** of dollars in weekly transaction volume, reducing payment failures by **40%** through idempotency improvements and retry logic overhaul.
- Redesigned order fulfillment microservice architecture, cutting infrastructure costs by **\$200K annually** while improving p99 latency by **35%**.
- Built seller onboarding and KYC verification workflows from scratch, reducing manual review backlog by **60%** within first quarter.
- Led technical interviews and grew engineering team by hiring **4 engineers** across backend and platform teams.
- Established on-call rotation and incident response protocols for marketplace platform serving **500K+** active users.

Staff Software Engineer/Manager, Google | 03/2015 - 04/2024

- Managed team of **5-7 engineers** building internal developer security platform used by **15K+** Google engineers for secrets management and access control.
- Built unified authentication gateway handling **billions** of daily requests across Google product surfaces, improving developer onboarding time by **50%**.
- Delivered Chrome-infra monitoring dashboard providing real-time visibility into browser update pipeline serving **hundreds of millions** of Chrome users.
- Designed and shipped certificate lifecycle management system replacing legacy PKI infrastructure, reducing certificate-related outages by **90%**.

- Mentored engineers on distributed systems design, resulting in **3 promotions** within team over 18-month period.

Software Engineer, Google | 02/2011 - 02/2015

- Built internal code review tooling adopted by **10K+** engineers, reducing code review turnaround time by **25%** through parallelization and caching improvements.
- Developed automated testing infrastructure for large-scale monorepo, cutting CI build times by **40%** via incremental test selection algorithms.
- Expanded team responsibilities to include frontend development using **Closure Templates** and **JavaScript** for internal UI components.
- Contributed to build system reliability, achieving **99.99%** uptime for critical developer tooling used company-wide.

Software Design Engineer II, Microsoft | 10/2007 - 12/2011

- Delivered reporting module for **Visual Studio** used by **thousands** of enterprise developers, shipping feature on schedule for Visual Studio 2010 release.
- Built data visualization components for **SQL Server Reporting Services**, improving report rendering performance by **30%** for large datasets.
- Owned reliability of build infrastructure processing **millions** of builds annually, reducing build failures by **15%** through improved error diagnostics.
- Participated in on-call rotation for production services supporting **Windows Azure** early adoption customers.

Software Design Engineer Intern, Microsoft | 06/2006 - 08/2006

- Delivered internal tooling prototype that was later productized, reducing team manual testing effort by **20 hours** per sprint.
- Implemented automated test framework for **.NET** configuration APIs, increasing test coverage from **60%** to **85%**.

Education

B.S. Honors Degree in Computer Science

Virginia Tech | 2004 - 2007